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Interactive effect of Fe and Mn deficiencies on physiological, biochemical, nutritional and growth status of soybean

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ABSTRACT

Iron (Fe) deficiency is one of the most common problems of soybean. It causes upper leaves yellowing, interveinal chlorosis, stunted growth and yield loss. Manganese (Mn) deficiency affects the reactions in the oxygen evolving complex (OEC) of photosystem II and increase the accumulation of reactive oxygen species (ROS). The aim of this research is to study the effect of Fe and Mn deficiencies applied separately and simultaneously on physiological, biochemical, nutritional and growth (morphological) parameters of soybean cultivars (Glycine max L.). The experiment was conducted in nutrient hydroponic solution lacking Fe or Mn or both Fe and Mn. Chlorophyll content index (CCI) and chlorophyll *a* fluorescence were measured through time to detect nutritional disorders at an early growth stage before the apparition of visual symptoms. The results showed that Fe and Mn deficiencies had a significant negative effect on the photosynthetic efficiency, CCI, stomatal conductance, protein content and shoot/root nutrient uptakes. Iron and manganese stress conditions were found to enhance the accumulation of secondary metabolites and increase the antioxidant activity such as total polyphenol content (TPC), malondialdehyde (MDA) and superoxide dismutase (SOD). These impacts were more accentuated when Fe and Mn stress were applied simultaneously than when any of the deficiencies was applied alone. More than that, Mn stress alone did not significantly affect the biomass accumulation. The obtained results showed that, in hydroponic conditions, iron and manganese rational fertilization can improve the studied parameters.

1. Introduction

Iron (Fe) is an essential plant micronutrient associated to the respiratory and the photosynthetic chains, Fe catalyzes reactions involved in metabolic process and required for several physiological functions, such as chlorophyll synthesis, photosynthesis, respiration, electron transport, enzyme activation and nitrogen fixation (Prasad, 2003; Santos et al., 2019). When Fe is not available to plants, these processes are disrupted at many levels and rates (Riaz and Guerinot., 2021; Aslani and Souri., 2018; Souri et al., 2018) affecting soybean's growth, development and yield. Iron deficiency chlorosis (IDC) is one of the most common problems of soybean (Hansen et al., 2003). The visual symptoms of IDC are interveinal chlorosis with dark green veins and reduced plant biomass. Iron is an important micronutrient that remains indispensable for the correct functioning of biological processes such as chlorophyll biosynthesis, photosynthesis (Jeong and Connolly., 2009). In plants, Fe serves as a cofactor for enzymes involved in photosynthesis, chlorophyll production, and the respiratory electron transport chain. (Schmidt et al., 2020).

Manganese (Mn) is another important micronutrient with a crucial physiological role. It is considered as an enzyme co-factor in oxygen evolving complex of photosystem II. In plants, Mn deficiency can occur without expressing visual leaf symptoms. However, if a critical low concentration is exceeded, interveinal chlorosis and yellow mottled leaves are the main visible symptoms (Schmidt et al., 2016) affecting the newly emerged leaves since Mn is not mobile in the plant's phloem (Loneragan, 1988). In extreme cases of deficiency, the accumulation of reactive oxygen species (ROS) increases, manifesting an oxidative stress symptom. This leads to brown spots which appear between the veins inducing leaf necrosis (Hannam., 1988). Mn is a vital microelement for plant growth and is involved in numerous biological processes, including the splitting of water molecules and the activation of enzymes. (Alejandro et al., 2020).

Cultivated soybean (Glycine max (L.) Merrill) is one of the most important pulse crops, ensuring nutritional and food security (Hailemariam et al., 2021; Johnson et al., 2022). It is known as a unique crop

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since it is rich in both proteins and lipids (Bellaloui and Kassem., 2021). Also, it is a good source of mineral nutrients and secondary metabolites, which are essential for human health. Compared to other species, soybeans have more isoflavones and higher secondary metabolite (Choi et al., 2022). Furthermore, it is the most leading economic and profitable vegetable oilseed crop in the world for low- and middle-income nations (Manavalan et al., 2009).

To face oxidative stress, plants generally accumulate secondary metabolites able to vary cellular metabolism (Ksouri et al., 2008). Phenolic compounds (Tato et al., 2013), malondialdehyde (MDA) (Shalata et al., 2001) and superoxide dismutase (SOD) (Lopez-Millan et al., 2013) play a major role in the adaptative reaction to iron deficiency through their chelating and antioxidant properties. Thus, the aim of this study is to describe the effect of Fe and Mn deficiencies and the possible interaction between them through physiological, biochemical, nutritional and growth (morphological) status of soybean plants.

2. Material and methods

2.1. Plant material and hydroponic system management

The study was carried out on soybean plants [Glycine max (L.) Merr.] cultivar (cv.) 'MN0808' in Plant physiology lab (32°13′ 10,77″ N 7° 53′ 23,38″) at University Mohammed VI polytechnic (UM6P) in Benguerir city, Morocco. This cultivar was developed and selected in the University of Minnesota. Seeds of soybean were sown in plastic trays containing commercial peat and irrigated daily with distilled water. After emergence (1 week after sowing), the uniformly and good growing seedlings were selected and transferred to plastic containers (40 cm*30 cm) filled with 4l of Hoagland and Arnon (1950) nutrient solution modified according to Cole et al. (2021) where the amount of N, P, K, S, Ca, Mg, Cl, Fe, Mn, Cu, Zn, Mo and B were respectively 210.10, 30.97, 234.59, 64.13, 200.39, 48.61, 0.64, 2.90, 0.50, 0.02, 0.05, 0,05, 0.51 ppm. The products used for each nutrient were: KH_2PO_4, KNO_3, $Ca(NO_3)_2.4H_2O$, $MgSO_4.7H_2O$, K2SO4, H_3BO_3, $CuSO_4.5H_2O$, $MnSO_4.H_2O$, $Na_2MoO_4.HO_2$, $ZnSO_4.H_2O$. To avoid any nutritional shock due to the transplantation, the hydroponic solution was introduced to the seedlings in a progressive way from 20 to 100%. In order to keep the plant in optimal condition, the nutrient solution was changed every 5 days and constantly aerated. The pH was regulated between 5.8 and 6. Soybean plants were grown in a 16/8 h (light/dark) photoperiod, a 26/20 °C (light/dark) temperature and a 70–80% relative humidity (RH). The experimental design was a completely randomized block with five treatments and three replicates, each replicate had 9 plants. Three levels of Fe (0, 1.45 and 2.9 ppm) were tested in interaction with two rates of Mn (0,25 and 0,50 ppm) for a growing period of 6 weeks. The choice of treatments is based on the stoichiometric method according to Fe and Mn requirements for soybean. Plants were exposed to the following treatments: T0: (0Fe-1Mn), T1: (1/2Fe-1/2Mn), T2: (1/2Fe-1Mn), T3: (1Fe-1/2Mn), T4: (1Fe-1Mn).

2.2. Physiological parameters

2.2.1. Fluorescence measurements

A portable fluorimeter (Handy PEA, plant efficiency analyzer, Hansatech Instruments Ltd, King's Lynn, United Kingdom) was used to measure Chlorophyll *a* fluorescence. The soybean leaves were kept in the dark for 20 min and illuminated for 1 s by an array of six light-emitting diodes (peak 650 nm) using a photon flux of 3500 mmol m^{-2} s^{-1}. The area of the sample surface was 4 mm^2.

To evaluate the effects of stress conditions on PSII and electron transport, chlorophyll fluorescence is a nondestructive, powerful and reliable biomarker (Strasser et al., 2000,; Kalaji and Loboda, 2007; Osório et al., 2014; Oukarroum et al., 2015). Starting from 25 days after sowing (DAS), the ChlFa was measured every week during the growing season. In order to assess the effect of iron and manganese on the physiological state of soybean plants and prevent the nutritional disorders before the apparition of visual symptoms, 27 measurements per treatment were taken on the right side of young fully mature leaves and the fluorescence transient (OJIP curve) was recorded during the first second of leaf illumination.

F_o and F_m correspond to the initial and maximum Chl a fluorescence. F_V corresponds to the maximum variable Chl fluorescence.

The photosynthetic performance index (PIABS) indicates the potential performance index for energy conservation from photons absorbed by PSII to the reduction of intersystem electron acceptors; PIABS = [go/(1 − go)].[jPo/(1 − jPo)].[yo/(1 − yo)] where go = 1−VJ is the fraction of electrons transported beyond QA- per exciton trapped by the reaction centers (RC) of PSII. QA is the primary quinone electron acceptor of photosystem II. jPo is the maximum quantum yield of primary photochemistry, it corresponds to the efficiency by which an absorbed photon will be trapped by PSII reaction centers (RC). jPo = (1 _ F_0/F_m) = FV/Fm. γo/(1−γo) is estimated by JIP-test as equal to the ratio of reaction centers and the absorbance (RC/ABS).

PItotal is the product PIABS and the probability that an electron can move from the reduced intersystem electron acceptors to the PSI end-electron-acceptors (Tsimilli-Michael, 2020): PItotal = PIABS * dRo/(1 _ dRo) where dRo = (1 − Vi)/(1 − Vj). The efficiency with which an electron can move from the reduced intersystem electron acceptors to the Photosystem I (PSI) end electron acceptors dRo (parameter related to electron transfer rate at PSI acceptor side). Vt is defined as relative variable Chl a fluorescence at time t corresponding to $(F_t - F_o)/(F_m - F_o)$ and this expression can be taken as a measure of the fraction of the primary quinone electron acceptor of PSII in its reduced state [QA-/QA (total)]. VI = $[(F_m - F_i)/(F_m - F_o)]$ is the relative contribution of the I-P phase, it is related to photosystem I content (Oukarroum et al., 2009).

The total driving force on absorption basis $DF_{total} = \log (PI_{total})$ was suggested to estimate the global driving force of processes evaluated by the corresponding PI_{total}.

2.2.2. Chlorophyll content index

To study the impact of different iron and manganese rates on the photosynthetic pigments, the CCI measurements were taken by Chlorophyll meter SPAD-502Plus (Konica Minolta Optics, Inc. Japan). For each treatment, 27 measurements were taken in the right side of mature young leaves every week starting from 25 DAS.

2.2.3. Stomatal conductance

After 7 weeks from sowing (47 DAS), the stomatal conductance was measured on 27 young and fully developed leaves per treatment using SC-1 leaf porometer (Decagon Devices, Inc. Pullman WA,USA). Inside the culture chamber, the ambient temperature was 25 ± 1 °C and the relative humidity was between 70 and 80%. The total photoperiod was 16h/day provided by LED light spots.

2.3. Biochemical analysis and antioxidant enzyme activities

2.3.1. Determination of proline content

Bates et al. (1973) method was used for measuring proline. 2 ml of 100% methanol were used to homogenize 100 mg of plant material; the supernatant was extracted after centrifuging the mixture at 5000 rpm for 20 min. In a test tube, 2 ml of the supernatant was combined with 2 ml of acid-ninhydrin and 2 ml of glacial acetic acid for a reaction that lasted an hour at 100 °C and stopped with an ice bath. The reaction mixture was extracted with 4 ml of toluene and vortexed for 15–20 s in a test tube stirrer. At a wavelength of 520 nm, the proline was measured by a spectrophotometer (FLUOstar Omega Microplate Reader, BMG LABTECH, Offenburg, Germany) using the upper phase (red complex). Using a standard curve, proline concentration on a fresh weight basis was computed according to the following formula: [((xg proline/ml x ml toluene)]/[(g sample)/5] = (x moles proline/g of fresh weight material.

To prepare acid-ninhydrin, 1.25 g of ninhydrin were added to 30 ml of glacial acetic acid and 20 ml of 6 M phosphoric acid. The mixture is

agitated until dissolved. The reagent remains stable for 24 h and kept cool (stored at 4 °C).

2.3.2. Determination of polyphenols content

The amount of total phenolic compounds was determined calorimetrically according to Taga et al. (1984). Gallic acid was used as a standard for the measurements. 50 mg of fresh leaves was dissolved in 1 mL of 100% methanol, centrifuged at 5000 rpm for 20 min 100 µl of the supernatant was combined with 2 ml of 2% Na2CO3 and after 2 min, 100 µl of Folin Ciocalteau reagent was added. The whole mixture is allowed to stand at room temperature for 30 min in the dark. By a spectrophotometer (FLUOstar Omega Microplate Reader, BMG LABTECH, Offenburg, Germany), the absorbance was measured at 750 nm. The results were expressed as milligrams gallic acid equivalents (GAE) per gram of fresh leaves.

2.3.3. Determination of soluble sugars content

Soluble sugars were measured by Dubois and al. (1951) method using glucose as standard where 25 mg of fresh leaves were added to 2 ml of 80% ethanol and centrifuged for 10 min at 10000 rpm. 200 µl of the supernatant was mixed with 200 µl of 5% phenol. The mixture is set at room temperature for 1 h. Then, 1 ml of sulphuric acid was added, and the resulting solution was vortexed. The absorbance was read at 490 nm wavelength on the UV spectrophotometer (FLUOstar Omega Microplate Reader, BMG LABTECH, Offenburg, Germany). The results were expressed as milligrams glucose per gram of fresh leaves.

2.3.4. Determination of protein content

The amount of protein was measured by Bradford (1976) method using bovine serum albumin (BSA) as standard. 100 mg of fresh leaves were homogenized in 2 ml of phosphate buffer (50 mM, pH = 6). The mixture was centrifuged at 10000 rpm for 10 min 10 µl of the resulting supernatant was added to 10 µl of distilled water and 200 µl of Bradfort reagent. The mixture was incubated for 1 min and read by the spectrophotometer (FLUOstar Omega Microplate Reader, BMG LABTECH, Offenburg, Germany) at a wavelength of 595 nm. The results were expressed as milligrams of BSA per gram of fresh leaves.

2.3.5. Malondialdehyde MDA

The membrane lipid peroxidation was assayed by Heath and Packer (1968) method based on complex formation of malondialdehyde (MDA). 50 mg of fresh leaves was homogenized in 2 ml of TCA 0.1% and centrifuged at 10,000 rpm for 15 min. In a test tube, 1 ml of the resulting supernatant was added to 4 ml TCA 20% containing TBA 0.5%, The mixture was heated in water bath at temperature of 95 °C for 30 min. Then, the reaction was stopped by an ice bath. The final mixture was centrifuged at 10,000 rpm for 10 min. The unspecific activity was corrected by A600nm − A532nm. The concentration of MDA was calculated by an extinction coefficient ε of 155 mM^{-1} cm^{-1}.

2.3.6. Superoxide dismutase

Superoxide dismutase (SOD) activity was measured by Beyer and Fridovich (1987) method. The reaction mixture contained 2.14 g/l of methionine, 83 mg/l of riboflavin, and 67 mg/l of nitro blue tetrazolium salt (NBT) dissolved in 50 mM sodium phosphate buffer (pH 7.8). 4 ml of the final mixture were added to 200 µl of enzyme extract. For 3 min, the samples were illuminated using LED (Ingelec; 30W 170–265V; 50/60Hz; I:0,27A) light spot. The mixtures were read at a wavelength of 560 nm against the blank. The results were expressed in U/mg protein where 1 U is the amount of change in the percentage of inhibition ($\%I = \frac{Ab-As}{Ab}$) by 50% and Ab is the blank absorbance and As is the sample absorbance.

2.4. Plant growth and mineral composition

2.4.1. Root's morphology

Plants root were cut carefully and washed with distilled water 47 DAS. They were spread over a plastic box, scanned using an Epson Perfection LA2400 scanner and the root images were digitalized and processed by the WinRHIZO image analyzing system (Regent Instruments Inc. Quebec, Canada). Root morphology measurements focused mainly on total root length, average root diameter, root surface area, root projected area and root volume.

2.4.2. Shoot and root biomass

47 DAS, the harvested plants were first washed with distilled water and root parts were immediately speared to shoot. We measured the shoot and root fresh weight, then we oven dried it at 70 °C for 3 days to determine biomass of dry weights.

2.4.3. Leaf area index

The leaves of each plant of the five treatments were put under a white background then pictured and analyzed using ImageJ (Version 1.53r) software. The measurements were taken at the end of the experiment (47 DAS)

2.4.4. Nutrients uptake

Dried shoot and root were powdered, digested by nitric acid (4 M HNO_3) and analyzed for macronutrient content (N, P, K and Mg) and micronutrient content (Fe, Mn, Zn and Cu) using Inductively Coupled Plasma Optical Emission Spectrometry (Agilent 5110 ICP-OES, USA).

2.5. Statistical analysis

The statistical analysis was conducted with SPSS data processing software (SPSS 20.0). The data were subjected to an analysis of variance (ANOVA). Means and standard errors were determined for each parameter and were evaluated by Tukey's HSD test at 0,05 probability level.

3. Results

3.1. Physiological parameters

3.1.1. Fluorescence OJIP and fluorescence parameters

The OJIP curves were used to understand the kinetic of chlorophyll fluorescence and to have an idea about the photosystem II efficiency. When the leaves were dark adapted, the fluorescence quantum yield was at its lowest level (O level) (Oukarroum et al., 2009; Stirbet and Govindjee, 2011). The reaction centers and the electron transport chain were at a resting state. As soon as the lightning took place, the electrons start migrating from PSII to plastoquinone pool (PQ) via QA et QB (Schansker et al., 2014). The J level of the fluorescence yield has been reached when most of the electrons in the reaction centers reduced the QA molecules. The thermal phase (I level) of fluorescence quantum yield is attained when QB molecules are similarly reduced. Fluorescence quantum yield (P level) reaches its highest at the same time as the PQ pool has reached its peak of reduction (Oukarroum et al., 2015; Hendrik et al., 2019). The increase of the fluorescence intensity in (I-P) was associated with the reduction of PSI transporters such as ferredoxin (fd) (Schreiber et al., 1989; Schansker et al., 2003) which is an iron sulfur protein. In less than 1 s, the polyphasic fluorescence transient rises from Fo to Fm, passing through Fv.

Fig. 1. showed 4 OJIP curves of soybean leaves subjected to different concentrations of iron and manganese. The measurements were taken at day 25 (a), 32 (b), 39 (c) and 47 (d) after sowing. The results obtained show that the differences between the treatments became more pronounced over time. Fig. 1.a illustrated that there were no visual differences between the 5 treatments at day 25 after sowing (DAS). As the

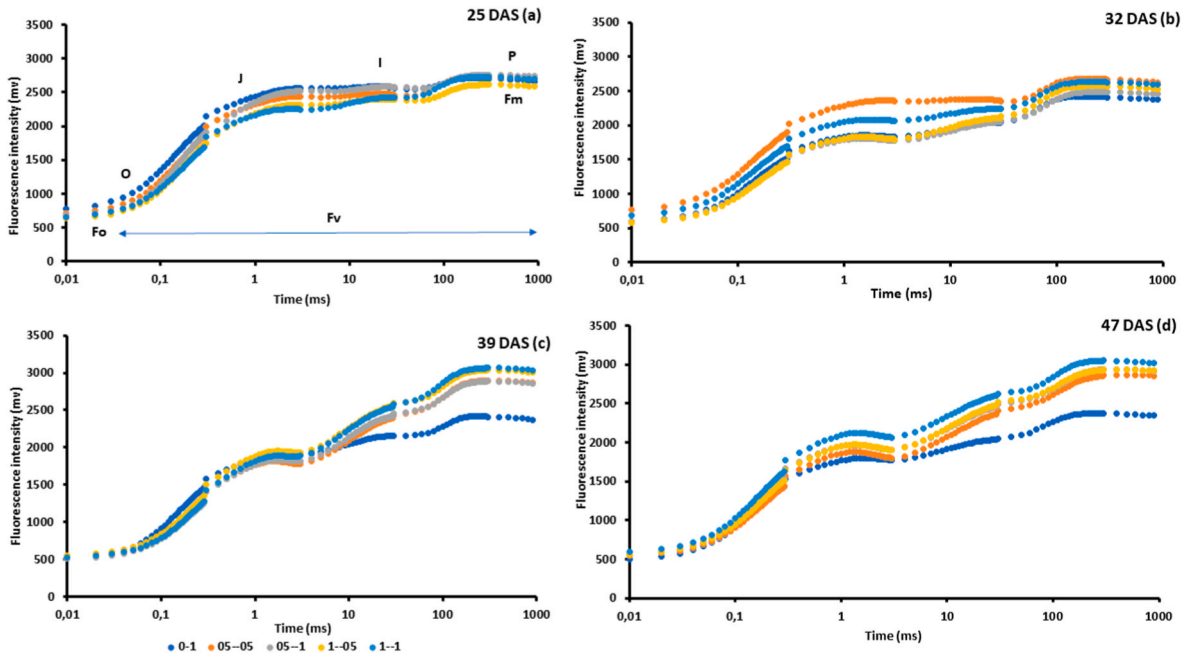


Fig. 1. The effects of Fe and Mn deficiencies on typical Chlorophyll *a* polyphasic fluorescence rise OJIP. The transients are plotted on a logarithmic time scale. The marks indicate the time points used by the JIP-test for the calculation of structural and functional parameters. The signals are the fluorescence intensity F_o (at 20 μs); the fluorescence intensities F_J (at 2 ms) and F_I (at 30 ms); the maximal fluorescence intensity, $F_P = F_M$. Each treatment is represented by the average of 27 measurements. (a), (b), (c) and (d) referred respectively to the measurements taken 25, 32, 39 and 47 DAS.

treatment goes, the degree of deficiencies increased gradually passing through Fig. 1b and Fig. 1c where the shape of I and P phases were distinguished. However, the O and J steps seemed to be unaffected. At the end of the experiment, 47 DAS, the iron and manganese concentrations had a significant effect on the shapes of OJIP transient especially in the JIP-step (Fig. 1d). T0: (0Fe-1Mn) had the lowest value of fluorescence yield followed by T1: (1/2Fe-1/2Mn). Then, T2: (1/2Fe-1Mn) and T3: (1Fe-1/2Mn) had a similar sharp of kinetic curves. T4: (1Fe-1Mn) control had the highest fluorescence yield.

The fluorescence parameters Fv/Fm, Fv/F0, PIabs, PItotal and DFtotal were presented respectively in Fig. 3b, Fig. 3c, Fig. 3d, Fig. 3e and Fig. 3f The results showed that iron and manganese deficiencies had a negative effect on these parameters. The statistical analysis revealed the presence of 4 distinct groups: the control T4 (group1) had the highest values followed by T2 and T3 which belonged to the same group (group2), then T1 (group3) and finally T0 (group4) which had the lowest values of each parameter.

3.1.2. *Chlorophyll content index (CCI)*

Same as chlorophyll fluorescence, chlorophyll content index (CCI)

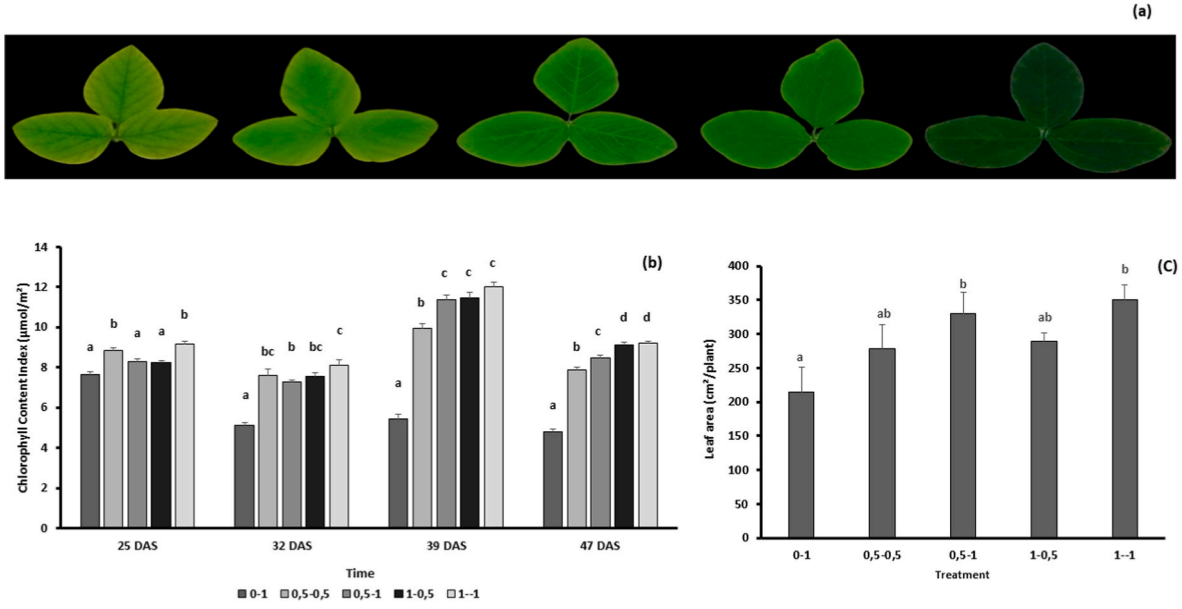


Fig. 2. Interactive effects of Fe and Mn deficiencies on color (a), Chlorophyll content index CCI (b) and leaf area (c) of soybean leaves. The pictures and the leaf area were taken at 47 DAS, the CCI was measured at 25, 32, 39 and 47 DAS. Values of the same day are means of 27 replicates ± Standard errors, different letters indicate significant differences at $p < 0.05$ according to Tuckey's test.

was measured every week to follow up the effect of the different treatments over time. Fig. 2b showed the 4 measurements of CCI. Iron and manganese deficiencies showed a negative effect on CCI and the increment between the 5 treatments was growing from the first to the last measure. The CCI had significatively decreased in T0 by 16%, 37%, 55% and 48% and had been reduced in T1 by 3%, 6%, 17% and 14% compared to control T4 respectively at 25, 32, 39 and 47 DAS compared to the control T4. However, the impact on T2 and T3 was less substantial.

3.1.3. Leaf area

Fig. 2.c showed that comparing to the control T4 where iron and manganese where fully available for soybean plants, the leaf area decreased by 39% for T0 which had no iron supply, by 21% for T1 which contained half of Fe and Mn crop requirements, by 18% for T3 which had half of Mn requirements and by 6% for T2 which had half of Fe crop requirements. To conclude, the lack of manganese impacted the leaf area more than iron deficiency.

3.1.4. Stomatal conductance

Nutrient translocation in plants may be indirectly explained by stomatal conductance. In this research, it was significantly affected by iron and manganese deficiencies at 47 DAS. Fig. 3a showed that iron had more impact on the rate of gas exchange and transpiration than manganese. Comparing to the control T4 (158,44 mmol/m^2.s); no iron application T0 caused a sharp increase of stomatal conductance by 42%. Almost the same rate of rise (34–35%) was recorded respectively in T2 where the iron supply was reduced to 50% and T1 where both Fe and Mn supplies were declined by 50%. An increase of 24% was recorded in T3 where only half of Mn supply was given to the plants.

3.2. Biochemical analysis and antioxidant enzyme activities

3.2.1. Total polyphenols content (TPC)

Fig. 4a represented the TPC differences between the five treatments. The lowest value corresponded to T0 with an average of 1,87 mg gallic acid/g of fresh leaf. The highest value was recorded in T1 (5,10), followed by T2 (3,26), then (2,85) and T4 (2,78). Statistical analysis proved that iron and manganese deficiencies had significant effect on total polyphenols content. The more severe the deficiencies, the higher is the value of TPC.

3.2.2. Total soluble sugars (TSS)

The obtained values of soluble sugars content varied from 1,57 to 1,72 mg glucose/g fresh leaf (Fig. 4b). However, the difference in TSS among treatments was not statistically significant ($p < 0,05$).

3.2.3. Proline content

Regarding average values of proline content, it was found that the results were approximately the same for all the treatments and balanced from 0,12 to 0,13 mg proline/g fresh leaf (Fig. 4c). The iron and manganese stress had no significant effect on the proline content.

3.2.4. Malondialdehyde MDA

MDA is generally used as a lipid peroxidation marker referring to the level of the oxidative damage of the cell membrane. Fig. 4d showed that decreasing iron and manganese supplies for soybean caused significant increases in MDA. T0 had an average value of 1,35 µmmol/g and was found to be the most affected treatment regarding lipid peroxidation in soybean leaves. Comparing to the control T4 with an average value of 0,75 µmmol/g fresh leaf, the level of MDA increased by 44% in plants received no iron, by 33%, 33%, 11% respectively for T1, T2 and T3.

3.2.5. Protein content

Fig. 4.e showed that iron and manganese deficiencies had a significant decrease on the leaf's protein content of soybean plants. The data illustrated that the control T4 had the highest average value (0,74 mg of BSA/g leaf fresh weight), followed by T3, T2, T1 then T0 which had the lowest value (0,56). The percentages of decreases were respectively 24%, 9%, 7%, 4% in response to Fe and Mn supplies respectively for T0, T1, T2 and T3 as compared to their control T4.

3.2.6. Superoxide dismutase (SOD)

SOD is one of the main reactive oxygen species (ROS) in the cell which serves as an important antioxidant. Fig. 4.f showed that superoxide dismutase activity increased gradually and significantly in

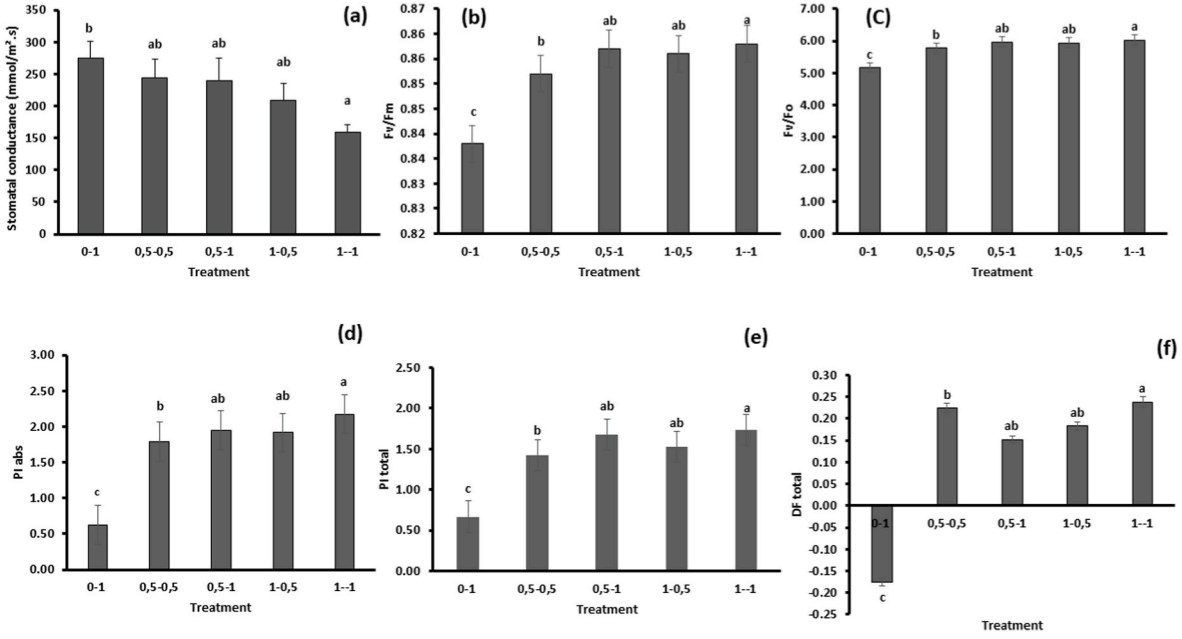


Fig. 3. The effect of iron and manganese stress on stomatal conductance of the soybean leaves. The measurements were taken 47 DAS. Mineral stress was applied 7 DAS for 6 weeks and each value represents the mean of 27 replicates ± Standard errors. Dissimilar letters indicate significant differences at $p < 0.05$ according to Tuckey's test.

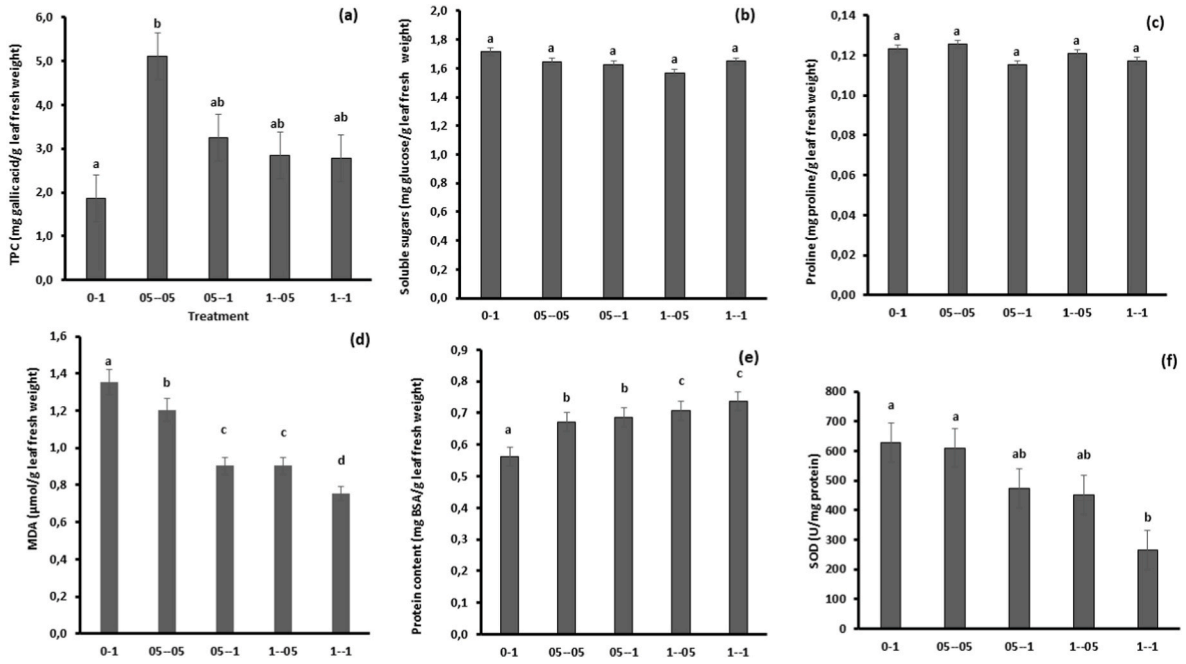


Fig. 4. Changes at 47 DAS in total polyphenols content TPC (a), total soluble sugars TSS (b), proline content (c), malondialdehyde MDA (d), protein content (e), superoxide dismutase activity SOD (f) in the leaves of soybean under iron and manganese deficiencies. Values are means of 3 replicates ± Standard errors, dissimilar letters indicate significant differences at $p < 0.05$ according to Tuckey's test.

soybean leaves grown under iron and manganese stress. Compared to the control T4 (264,46 U/mg protein), SOD activity increased it by 238% for T0 (628,91 U/mg protein), by 231% for T1(230,96 U/mg protein), by 179% for T3 (474,70 U/mg protein) and by 171% for T2 (451,76 U/mg protein).

3.3. Plant growth and mineral composition

3.3.1. Roots morphology

Changes in root morphology parameters in soybean plants grown under iron and manganese stress are presented in Fig. 5. Fe and Mn deficiencies affected visually and significantly the roots morphology (Fig. 5a). The lowest values of the root parameters such as length (Fig. 5b), surface (Fig. 5d), volume (Fig. 5c) and tips (Fig. 5e) corresponded to T0 followed by T1 with a respective decrease of 46% and

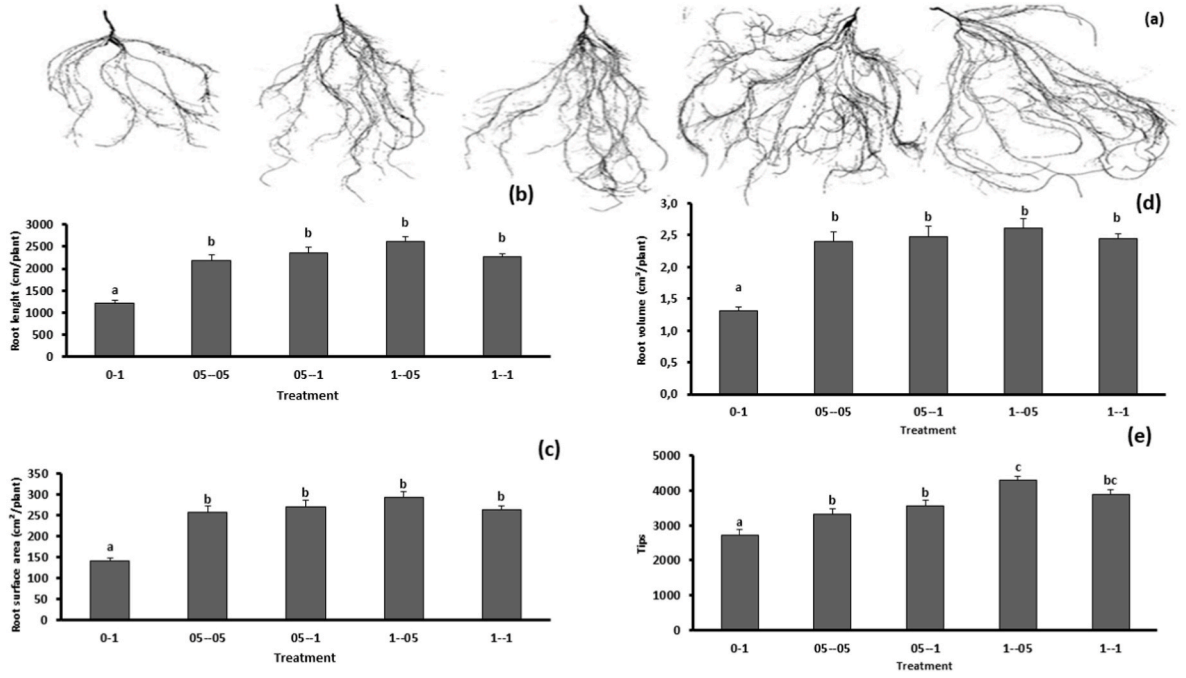


Fig. 5. Interactive effects of Fe and Mn stress on root morphology (a), root length (b), root surface area (c), root volume (d) and tips at 47 DAS. Each value represents the mean of 27 replicates ± Standard errors. Dissimilar letters indicate significant differences at $p < 0.05$ according to Tuckey's test.

3,9% compared to the control roots T4. However, the control had not the highest values of root's morphology, it has been exceeded by T3 with an average of 10% for all the parameters. This may explain that Mn deficiency can cause a root extension.

3.3.2. Leaf, stem and root biomass

Exposure of soybean plants to iron and manganese stress decreased significantly the fresh and dry biomasses (Fig. 6.). Compared to the control plants T4 and respectively for T0, T1, T2 and T3; the leaves biomasses decreased by 41%, 14%, 7% and 2% for the fresh weight (Fig. 6a) and by 38%, 25%, 14% and 10% for the dry weight (Fig. 6b); the stems biomasses drop by 42%, 18%, 11% and 4% (Fig. 6c) for the fresh weight and by 45%, 26%, 17% and 12% (Fig. 6d) for the dry weights. The root's fresh weights (Fig. 6e) confirmed the analyzes done by Winrhizo: besides T0 which had the lowest value, there was no significant differences between the other treatments. For the root's dry weights presented in Fig. 6.f, T3: (1Fe-1/2Mn) had the highest value of 0,16 g/plant, followed by T2: (1/2Fe-1Mn) and T4: (1Fe-1Mn) which had the same dry weight, then T1: (1/2Fe-1/2Mn), and finally T0: (0Fe-1Mn) with 0,10g/plant.

3.3.3. Fe and Mn shoot and root uptakes

Iron and Manganese shoot and root uptakes were significantly and gradually affected by Fe and Mn supplies. Compared T0 to T2, increasing iron supply from 0 to 1,45 ppm allowed to increase Fe shoot uptake by 35% and Fe root uptakes by 90% and increasing it from 1,45 ppm (T2) to 2,9 ppm (T4) improved Fe shoot uptake by 14% and Fe root uptake by 33% (Fig. 7a). Manganese shoot and root uptakes were improved respectively by 34% and 37% once the manganese supply was increased from 0,25 (T3) to 0,5 ppm (T4) (Fig. 7b).

3.3.4. Phosphorus (P), potassium (K) and magnesium (Mg) shoot and root uptakes

Chemical analysis showed that iron and manganese supplies had an interactive effect with P-root uptake, K shoot and root uptakes and Mg shoot uptake. A negative interactive effect was observed for P and K, the higher the concentration of iron and manganese in the hydroponic solution, the lower the phosphorus (Fig. 7c) and potassium composition of soybean roots (Fig. 7d). On the contrary, the interactive effect was positive for magnesium; Mg-shoot concentrations were increased for treatments with no iron supply (Fig. 7e).

3.4. Principal component analysis (PCA) and correlation analysis

Principal component analysis (PCA) and correlation analysis were studied to find out the degree of divergence between the different parameters and to gather them into relatively homogeneous groups. It also calculates the contribution of each variable to the total variation between the principal axes of differentiation. Eigenvalues are often used to select the factors that best discriminate the inputs. The sum of the eigenvalues is usually equal to the number of variables. Fig. 8 and Fig. S2 summarized the relationships between the 42 measured parameters. In this analysis, the first and second principal components retained 23.58 and 8.92 of the original variables account for 63.75%, 24.11% of the total variation respectively. suggesting that they are the principal components and should be used to summarize the original 42 variables. Based on the two principal components, we found four homogeneous groups. Fig. 8 showed that group 1 is the largest, it includes 24 parameters (Chl 1:CCI 1; Chl2:CCI; Chl3:CCI3; Chl4:CCI4; Fv/Fm; Fv/Fo; PI abs; PI total; DF total; RS: Root surface; RL: Root length, RV: Root volume; T:Tips; LA: Leaf area; LFW: Leave's fresh weight; LDW: Leave's dry weight; SFW: Stem fresh weight; SDW: Stem dry weight; RFW: Root fresh weight; TPC: Total polyphenols content; PC: Protein content; FeS: Shoot-Iron; FeR: Root-Iron; FeT: Total-Iron). Group 2, Group 3 and group 4 included respectively 4 (MDA: Malondialdehyde; SC: Stomatal conductance; SOD: Superoxide dismutase; Pr: Proline), 5 (MnS: Shoot-Manganese; MnR; Root-Manganese; MnT: Total-Manganese; MgS: Shoot-Magnesium; SS: Soluble sugars) and 6 (MgR: Root-Magnesium; MgT: Total-Magnesium; PT: Total-phosphorus; KT: Total-potassium; PR: Root-phosphorus; KR: Root-Potassium) parameters. Component 1 had a high positive loading on group 1 and a high negative loading on group 2 and 3. Component 2 had a positive loading on group 2 and 4 and a negative one on group 3. Fig. S2 also showed that group 1 is negatively correlated with group 2 and 3. And groups 3 and 4 were negatively correlated with group 2.

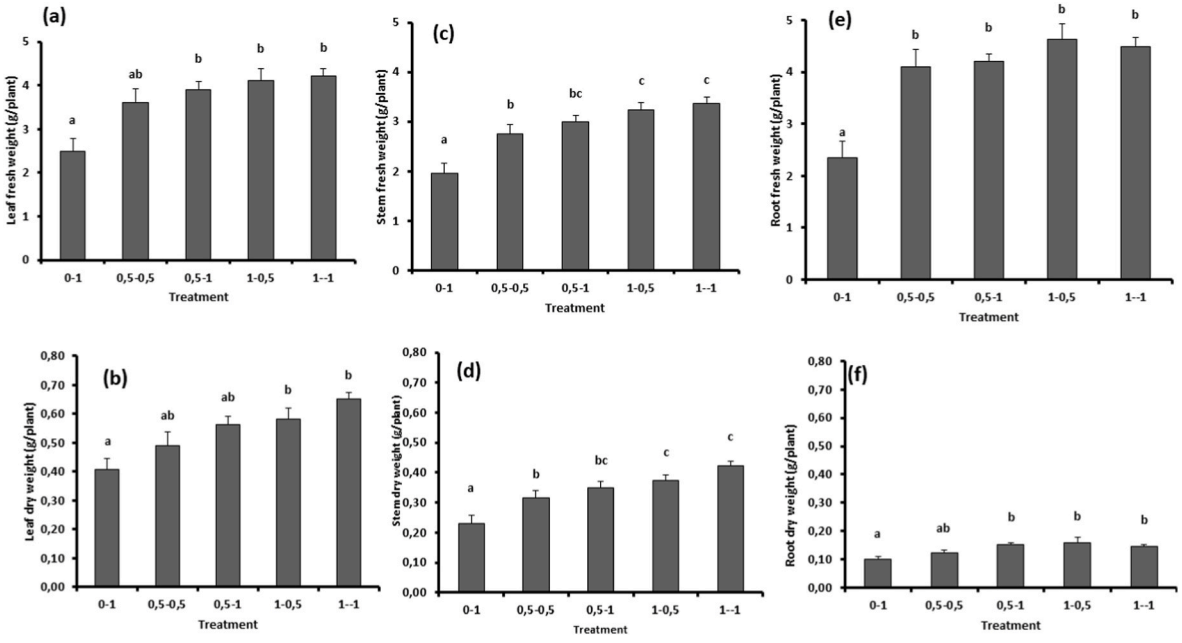


Fig. 6. The effect of Fe and Mn mineral stress on leaf fresh (a) and dry (b) weight, stem fresh (c) and dry (d) weight, root fresh (e) and dry (f) weight of soybean plants at 47 DAS. Each value represents the mean of 27 replicates ± Standard errors. Dissimilar letters indicate significant differences at $p < 0.05$ according to Tuckey's test.

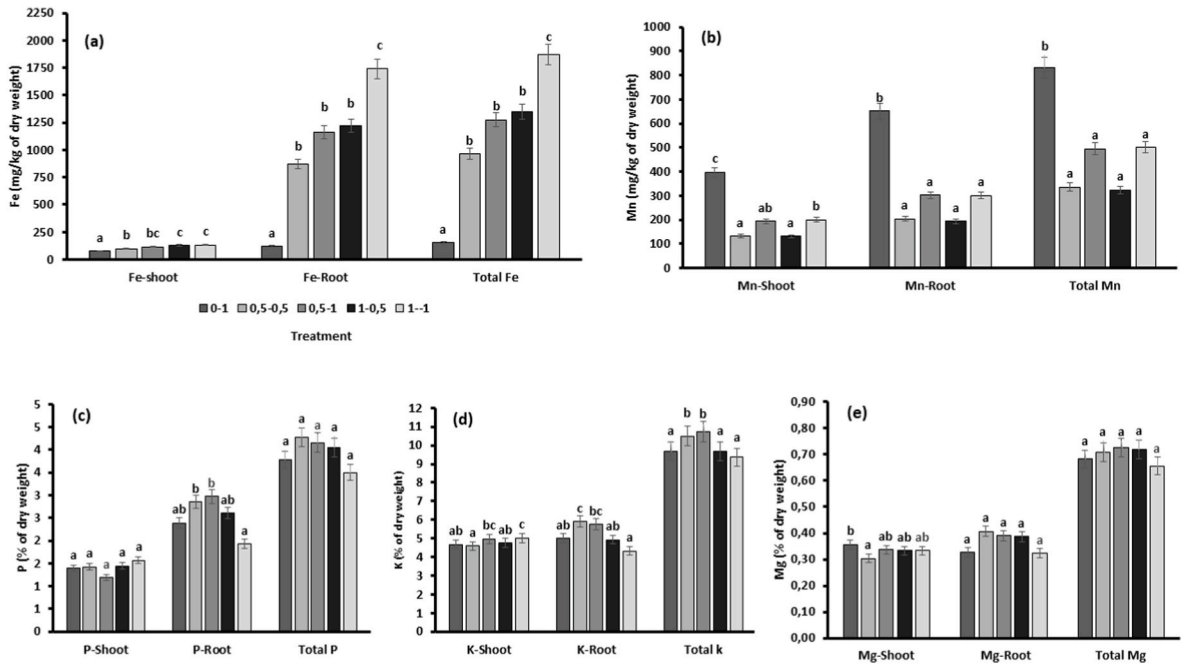


Fig. 7. Changes in shoot and root nutrients uptakes: Fe (a), Mn (b), P(c), K(d) et Mg(e) under Fe and Mn deficiencies. Each value represents the mean of 3 replicates ± Standard errors. Dissimilar letters indicate significant differences at $p < 0.05$ according to Tuckey's test.

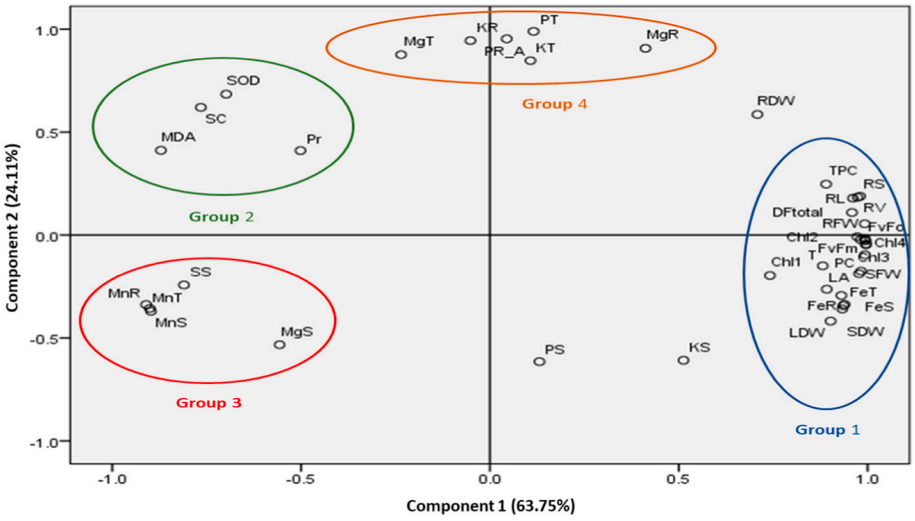


Fig. 8. Principal component analysis (PCA) of 42 measured parameters (group 1: 22 parameters; group 2: 4 parameters; group 3: 5 parameters; group 4: 6 parameters). Chl 1:CCI 1; Chl2:CCI; Chl3:CCI3; Chl4: CCI4; Fv/Fm; Fv/Fo; PI abs; PI total; DF total; RS: Root surface; RL: Root length, RV: Root volume; T: Tips; SC: Stomatal conductance; LA: Leaf area; LFW: Leave's fresh weight; LDW: Leave's dry weight; SFW: Stem fresh weight; SDW: Stem dry weight; RFW: Root fresh weight; RDW: Root dry weight; TPC: Total polyphenols content, Pr: Proline; PC: Protein content; MDA: Malondialdehyde; SS:Soluble sugars; SOD:Superoxide dismutase; PS:Shoot-phosporus; PR: Root-phosphorus; PT: Total-phosphorus; KS: Shoot-potassium; KR: Root-Potassium; KT: Total-potassium; MgS: Shoot-Magnesium; MgR: Root-Magnesium; MgT: Total-Magnesium; FeS: Shoot-Iron; FeR: Root-Iron; FeT:Total-Iron; MnS: Shoot-Manganese; MnR; Root-Manganese; MnT: Total-Manganese.

4. Discussion

In this experiment, we tried to analyze the effects of iron and manganese deficiencies on the photosynthetic apparatus by recording chlorophylla fluorescence of soybean's leaves. According to OJIP curve, the reduction of chlorophyll fluorescence is based on the reduction of quinone and plastoquinone PQ capacity (Strasser et al., 2000; Kalaji and Loboda, 2007). The induced stress of Fe and Mn caused a decrease of variable (Fv) and maximal (Fm) fluorescences compared to the control T4 (Fig. 3b and 3c). The same results were found by Donnini et al. (2013). This reduction occurred because of the impairment of PSII and PSI functions and the damage of PSII reaction centers (Roháček and Barták, 1999; Kalaji and Loboda, 2007; Breštič et al., 2015). On one hand, iron influences the photosynthetic processes at different levels since it is present in various forms such as Fe-S clusters in PSI, heme-bridged Fe in cytochrome *b*559 and non heme Fe in PSII. Fe participates in electron transport and catalyzes the redox reactions (Barker and Pilbeam, 2015; Roosta et al., 2018). Therefore, iron deficiency is responsible for the low rates of photochemical efficiency (Fig. 1), chlorophyll concentrations, chloroplast proteins (Morales et al., 1991), and PSII reaction center's efficiency reducing the stability of thylakoid membrane (Wang et al., 2022). Interveinal chlorosis is the visual symptom of iron deficiency caused by the reduction of chlorophyll content in the leaves (Fig. 2a) (Abadia et al., 1999). Iron deficiency chlorosis of soybean is described by interveinal chlorosis, stunted growth (Fig. 6) and development leading to yield reduction (Merry et al. 2021). Since the phloem mobility of Fe is low in plants, the symptoms are affecting mainly the young upper leaves (Fig. 2a). Under iron deficiency stress conditions, chlorophyll content (Fig. 2b) and carotenoidsin leaves are decreased, although only minor decreases are seen for carotenoids (Abadia et al., 1999). On the other hand, Manganese is an essential catalyst that affects the reactions in the oxygen evolving complex (OEC) of photosystem II (Vogt et al., 2015). During the light reactions of photosynthesis where the solar energy is converted to

chemical energy, the photo-oxidation of H_2O into O_2, electrons and protons cannot occur without the oxobridged cluster formed by manganese and calcium Mn_4CaO_5 (Satadal et al., 2017). The lack of manganese complexes produces a negative effect upon the integration and the stabilization of PSII which in turn affect the photosynthesis rate (Schmidt et al., 2016) and the thylakoid structure promoting chlorophyll degradation (Fig. 2a) and leaf chlorosis (Papadakis et al., 2007). Since Mn is not mobile in plants (Li et al., 2017; Ferrandon and Chamel, 1988), the symptoms of deficiency are observed on the young upper leaves (Fig. 2a). Mn deficiency reduces the number of chloroplasts and decreases net photosynthetic efficiency, lowering accordingly the chlorophyll content (Fig. 2b) (Henriques, 2004; Hebbern et al., 2009; Alejandro et al., 2017). This leads in turn to impaired growth and biomass loss (Fig. 6) (Longnecker et al., 1991; Hebbern et al., 2005; Pedas et al., 2005). In roots, Mn deficiency can cause an increase of root hair number and weight (Yang et al., 2013). Meanwhile, iron deficiency causes a poor root formation (Kobayachi et al., 2003). This was manifested in this research (Figs. 5 and 6.c and Fig. 6f) by the low values of root length (Fig. 5b), surface (Fig. 5c), volume (Fig. 5d) and number of tips (Fig. 5e) of T0, T1 and T2 where iron was deficient and the highest value of the same parameters of T3 (exceeding the control T4) where Manganese was deficient (Fig. 5a).

Fe deficiency increases Mn-SOD because free iron ions cause ROS formation (Lopez-Millan et al., 2013). Therefore, iron deficiency in T0, T1 and T2 caused high levels on SOD compared to T3 where Manganese was deficient. Under manganese stress, this damage is visualized by necrotic spots due to high level of free radicals in chloroplasts and low level of Mn-SOD activity (Fig. 4f) (Hajiboland, 2012; Broadley et al., 2012). Similar results were found by Allen et al. (2007) where Mn-SOD mitochondrial function (Fig. 4f) and photosynthesis rate (Figs. 1 and 2b) were decreased under Mn deficiency. The detoxification of reactive oxygen species (ROS) can be caused by enzymatic and non-enzymatic mechanisms which leads to oxidative damage to some cellular components such as membrane lipids, proteins and nucleic acids (Shalata et al., 2001; Blokhina et al., 2003). The same results were recorded in Fig. 4.d and Fig. 4.e where lipid peroxidation marker MDA increased, and protein content decreased under iron and manganese stress. A low iron intake lowers the amounts of soluble carbohydrates and proteins in cabbage. This may be because iron plays a part in the synthesis of sucrose and acts as a stimulator of sucrose phosphate synthase activity. As a result, iron deficiency inhibited plants from producing protein and sugar. (Wang et al., 2022).

Under Fe deficiency, plants accumulate H_2O_2 and superoxide anion radicals due to the consumption of antioxidant proteins such as ascorbate peroxidase, catalase, superoxide dismutase (Fig. 4f), and component of the electron transport system of chloroplasts and mitochondria (Tewari et al., 2015). The accumulation of H_2O_2 in the guard cells has been involved in stomatal closure (Fig. 3a) (Xiao Zhang et al., 2001; Bright et al., 2006). Also it was proved that stomatal conductance, transpiration rate, and other gas exchange characteristics can all be impacted by an iron deficiency (Higuchi and Saito, 2022). Chlorotic leaves caused reduction of the diameter of xylem vessels and impeded xylem conductivity (Eichert et al., 2010), this leads to the low transpiration rates and the decrease in stomatal opening (Fernández et al., 2008).

As a result of biotic and abiotic stress including iron deficiency, the biosynthesis and accumulation of polyphenols are stimulated (Naczk and Shahidi, 2004). This high concentration of TPC leads to a restriction (Fig. 6) in biomass production (Sreenivasulu et al., 2002). To resist to abiotic stresses, plants increase the contents and quality of TPC to improve the antioxidant capacity (Lin et al., 2006). In this research, Total phenolic content was higher in T1, T2, T3 comparing to the control T4 (Fig. 4a). However, TPC in T0 leaves which had no iron supply at all had the lowest average value. This result should be more studied in further studies. According to Giorgi et al. (2009), the phenylalanine ammonialyase (PAL) enzyme is stimulated by the accumulation of TPC under iron deficiency. Furthermore, PAL catalyze the transformation of L-phenyl alanine into trans-cinnamic acid, which is the prime intermediary in phenolics biosynthesis. Also, Mn can be used as cofactor of PAL (Engelsma, 1972; Alejandro et al., 2020) which explains more the effect of the interaction between iron and manganese deficiencies on TPC.

Nutrient availability determines plant's growth and development. It is important to understand the dynamics, assimilation, transport, and biological interactions between the different nutrients to ensure a good productivity of crop plants (Wawrzyńska and Sirko, 2014). Fe- and Mn-deficiencies influenced phosphorus (P) root uptake. There are some studies that pointed out that P and Fe nutrition have similar physiological connections (Zheng et al., 2009). Under Fe deficiency, P acquisition in shoots and roots is stimulated, while in contrary, P deficiency greatly enhances Fe availability within the plants (Misson et al., 2005; Briat et al., 2015). The transcription of P-related genes is altered by Fe deficiency (Moran et al., 2014). This confirmed the results found in Fig. 7.a and Fig. 7.c.

Author contributions

BEA and AO conceptualized and designed the experiment and lab studies. BEA and RN performed the studies. BEA, RN, FM analyzed the samples and data. BEA, FM wrote the paper. All authors contributed to the article and approved the submitted version.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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